MILITARY SPECIFICATION

Test Set, Fire Control System AN\AWM-87

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - equipment covered by this specification, is intended for organizational maintainance use. This equipment, together with auxiliary ground support equipment and built-in-test (BIT) operational check, will provide a check of the Radar Set AN\APQ-72D, Radar Set Group AN/APA-170A, and Target Intercept (Dogfight) Computer CP-1065\A.

1.2 Classification - The equipment covered by this specification shall consist of the following items:

Item	Drawing or Model Number	Nomenclature	Reference Paragraph
1	50-014258	Cable Set, Interconnecting Cable Group	3.6.1, 3.7.1
2	50-014000	Fire Control System Test Set TS-3649/AWM-87	3.6.2, 3.7.2
3	50-014001	Electrical Cable Test Set TS-3648/AWM-87	3.6.3, 3.7.3

1.3 Associated Equipment - The equipment shall operate with the following associated equipment:

HD 747/AWM-22 or AN/AWA-6 Air Hydraulic Cart

COMMENTS OR RECOMENDATIONS WHICH MAY BE OF USE IN IMPROVING THIS DOCUMENT MAY BE SENT TO THE PREPARING ACTIVITY WHEN A SELF-ADDRESSED STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL (DD FORM 1426) APPEARS AT THE END OF THIS DOCUMENT, OR BY LETTER ADDRESSED TO THE COGNIZANT ACTIVITY

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9 APPLICABLE DOCUMENTS

2.1 General- The current issue of the following documents at the time of contract award form a part of this specification to the extend specified herein:

BULLETINS

Military	
MIL 400	Airborne Electronic and Associated Equipment
MIL-E-17555	Electronic and Electrical Equipment Accessories, and Repair Parts, Packaging and Packing of
MIL-T-18303	Test Procedures, Reproduction Acceptance and Life for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical and Aeronautical Support Equipment Including Ground Support Equipment

HANDBOOKS

Military	
MIL 235-2 Navy	Electromagnetic (Radiated) Environmental Consideration for Design and Procurement of Electrical and Electronic Equipment

SPECIFICATIONS

MIL-T-21200	Test Equipment for use with Electronic and Electrical Equipment, General Specification for
MIL-I-45208	inspection System Requirements
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys

STANDARDS

Military	
MIL-STD-143	Standards and Specifications, order of Precedence for the Selection of
MIL-STD-461	Interface Characteristics Requirements for Equipment

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MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-471	Maintainabillty Demonstration
MIL-STD-704	Aircraft, Electrical Power Characteristics
MIL-STD-781	Reliability Test, Exponential Distribution
MIL-STD-810	Environmental Test Methods

2.1.1 Availability of Documents - When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

2.2 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

a. Contract - The contract shall have precedence over any specifications.

b. This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

c. Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extend specified.

3. REQUIREMENTS

3.1 Testing - This specification makes provision for testing (see Quality Assurance Provisions, paragraph 4.).

3.2 Parts, Materials and Processes - Parts, materials and processes shall be .selected as required to satisfy the total requirements of this specification. Microelectronic devices shall be given first consideration in design. Parts, materials and processes shall conform to applicable specifications as specified in Military Bulletin 400,

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except as specified herein.

Parts, materials, and processes covered by the specifications and publications listed in military Bulletin 400 shall be considered as standard and shall be used whenever they are suitable for the purpose. Nonstandard parts, materials, and processes must be equivalent to or better than similar standard parts, materials, and processes. When Military Bulletin 400 fails to provide an applicable specification or standard, the contractor shall use other established specifications or standards in the order of precedence set forth in MTL-STD-143. Parts, materials, and processes thus selected from other than Military Bulletin 400 shall not be considered as standard, and approval must be obtained prior to their use in equipment.

3.3 Design and Construction - The design and construction of the equipment shall be in accordance with MIL-T-21200, as modified herein.

3.3.1 Total Weight - The total weight of the equipment, including cables, shall be a minimum consistent with good design and shall not exceed 150 pounds.

3.3.2 Modules, Maintenance - The electronic portions of the equipment shall be divided into maintenance modules where possible. Modules shall normally be considered repairable.

3.3.3 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with MIL-STD-454, requirement 10. Insofar as is practicable, all connectors on the front panel shall be mounted along the lower edge of the panel and in no case shall their location be such that the connecting cables will interfere with the operating controls.

3.3.3.1 Cabling - The equipment cabeling shall be compatible with the equipment to be tested. It shall be the electrical equivalent of the cabling used in the normal installation of the equipment being tested, per MIL-STD-454, requirement 66.

3.3.4 Human Engineeringg - Human engineering shall be in accordance with MIL-STD-454, requirement 62.

3.4 Availabity - The availability shall be a minimum of 99.0 percent when calculated using the results of the Reliability and Maintainability y demonstrations of paragraphs 3.4.1 and 3.4.2 as fallows:

$$\text{Availability} = \frac{\Theta}{\Theta + M_{ct}}$$

where:

0 is the observed Mean-Time- Between-Failure (MTBF) equal to the total operating time of the equipment divided by the number of failures, as defined in MIL-STD-781.

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and:

M_{ct} is equal to the Mean Corrective maintenance downtime, as defined in MIL-STD-471

3.4.1 Reliability - The contractor shall conduct a reliability program in accordance with MIL-STD-454, requirement 35.

3.4.1.1 Operating Stability - The test equipment shall perform as defined herein when operated continuously or intermittently for periods of nine months minimum without necessity for readjustment of any controls not readily accessible to the operator during normal use, excluding repair actions.

3.4.1.2 Operating Life - The equipment shall have a minimum total operating life of 2,000 hours. Parts requiring scheduled replacement due to wear during the life of the equipment and, the wearout life of such parts, shall be determined by the contractor and submitted to the procuring activity.

3.4.1.3 Reliability in Mean-Time-Between- Failure (MTBF) - The equipment, including any built-in-test provisions, shall have an MTBF of 500 hours.

3.4.2 Maintainability - The contractor shall conduct a maintainability program as required by the contract. The program plan shall be approved by the procuring activity. Built-in-test features, modular construction and packaging, and provisions for test points shall be utilized as appropriate.

3.4.2.1 Maintenance Times - The equipment, Including any built-in provisions, shall be designed to meet the following quantitative maintainability requirements for corrective maintenance when tested in accordance with MIL-STD-471.

a. Mean time to repair, M_{ct} 2 hours

b. Maximum time to repair, $M_{max\ ct}$ 7.6 hours (at the 90th percentile).

3.4.2.2 Accessibility - The equipment shall be assembled as a fully functional structure attached to the control panel and operational when removed from the combination case except that EMI requirements shall not apply when outside the case.

Plug-in printed circuit assemblies and modules shall be used where practical for active functional circuitry and these shall be arranged for ease of removal and replacement.

3.4.3 Interference Control - The generation of electromagnetic interference by the equipment and the vulnerability of the equipment to electromagnetic interference shall be cotrolled Within the limits of MIL-STD-461 unless otherwise specified. The equipment subject to and comply with the tests of MIL-HBK-235-2, Navy, Table 5A.

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3.4.4 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-STD-454, requirement 34 and MIL-N-18307.

3.4.5 Service Conditions - Service conditions (environmental) shall be:

3.4.5.1	Altitude	
	Non-Operating	50,000 ft maximum
	Operating	10,000 ft maximum
3.4.5.2	Shock	The shock pulse shape shall be half sine wave with a peak amplitude of 20 g's and a nominal time of II milli-seconds.
3.4.5.3	Vibration	
	Operating	5 Hz to 15 HZ at 0.06 " ± 0. 012" peak-to-peak 16 Hz to 25 Hz at 0.04 ± 0.008" peak-to-peak 26 Hz to 55 Hz at O. 02 " ± 0.004" peak-to-peak
3.4.5.4	Humidity	
	Non-Operating	Cover removed, 71°C(+159°F), 95% R. H.
	Operating	Cover removed, 51. 5°(+124°F) , 85% R.H.
3.4.5.5	Salt Atmosphere	Cover removed, 35°C(+9@F), 5% 48 hours
3.4.5.6	Sand and Dust	75% 325 mesh, 12 hours, 1750 f_{pm}
3.4.5.7	Explosive Atmosphere	MIL-STD-810, Method 511, Procedure 1
3.4.5.8	High Temperature	
	Non-Operating	+85°C (+185°F)
	Operating	+55°C (+131°F)

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3.4.5.9 Low Temperature

Non-operating	-40°C (-40°F)
Operating	-40°c (-40°W

3.4.5.10 Acoustic environment - The equipment shall operate within the specifications of this document in an acoustic noise level of 100 dB above a reference level of 0.0002 microbars. The power distribution shall be continuous from 20 Hz to 9600 Hz and shall be flat (-2, +4 dB).

3.4.6 Standard Conditions - The following conditions Shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room Ambient
Altitude	Normal Ground
Vibration	None
Humidity	Room Ambient
Input Power Voltage	The equipment shall operate as specified herein when powered from a single phase, 115 VAC, 400 Hz power source as defined in MIL-T-21200 for military 400 Hz power. Maximum power shall be less than 200 volt amperes per unit.

3.5 Transportability - Transportability requirements shall be in accordance with MIL-T-21200.

3.6 Performance - The performance of the equipment shall be as specified herein.

3.6.1 Cable Set, Interconnecting Cable Group - A set of three interconnect cables shall be provided. These cables shall connect the equipment to the aircraft as follows:

AWM- 87 Attachment	Cable Designation	Aircraft Attachment
W12J2	W12	J112 (73-J404)
W13Jl	W13	J113 (73-J405)
W5Jl	W5	Umbilicals Plus Motor Fire Connector

3.6.2 Test Set, Fire Control System TS-3649/AWM-87- Operational status of the radar set AN/APQ-72D, Radar Set Group A N/APA-170A and Target Intercept (Dogfight) Computer CP-1065A, shall be measured with the Fire Control System Test Set (FCSTS). The

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FCSTS provides inputs to the F-4N AERO IA Airborne Missle Control System (AMCS) and measures the output parameters generated by the AMCS. The equipment provider the proper loading and current capacity for all input and output signals. The signals provided to the AMCS are ramp voltages from 0 to ± 100 VDC and voltages from -100 VDC to + 100 VDC with a tolerance of +0.75%= The AMCS output signals measured by the equipment are:

Voltage (DC)	-250 to +300
Voltage (AC)	0 to 125
Current (ma)	100 to 150
Time (see)	0 to 30

Tolerance of all output values is ± 1%

3.6.2.1 Built-In Test - The equipment shall contain integral circuitry activated by front panel controls that generate the necessary test signals and references to enable end-to-end functional checks. This is accomplished in a self test parity check, a lamp test and self check. The confidence level of the built-in test shall be 90% minimum.

3.6.2.2 Operating Modes - The FCSTS shall have a front panel toggle switch for selection of SELF TEST, LAMP TEST or NORMAL.

3. 6.2.3 Measurement Capabilities - In the NORMAL mode, the FCSTS can be sequentially stepped through various groups of tester functions by use of the function select and increment switches. Actual test parameter value shall be displayed digitally. Parameter acceptance shall be indicated by a f rent panel GO indicator and rejection shall be indicated by a NO-G0 indicator to the intent specified. The following tests will be performed with the FCSTS automatically programming the appropriate input lines. Parameters shall be measured with the values indicated through automatic selection of the appropriate reading line. An OPERATOR ACTION light will illuminate when a change in the aircraft settings are required. An OPERATOR OBSV light will illuminate when an aircraft function result is required. Measurement accuracy of all nominal values shall be ± 1 % of measured value.

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TESTER FUNCTION	SPECIFICATION		MAXIMUM LOADING
3.6.2.3.1			
CW TRANSMITTING AND MODULATION CHECK			
RADAR STBY MCP POWER ON	165 ± 35 VDC	W13P1-C	l MEG
KLYSTRON MAG CURRENT	130 ± MA	W13P1 P/E	95k/95k
30 VDC DELAYED	ON WITHIN 5 ± 0.3 SEC	W13P1-q	‘1 MEG
RADAR +300V	300+3 VDC	W13P1-AB	1 MEG
RADAR +250V	250+2.5 VDC	W13P1- +	1 MEG
RADAR 150V	150+1.5 VDC	w13P1- N	1 MEG
RADAR -250V	-350.2.5 VDC	W13P1-c	1 MEG
AIM-2 ( ) RANGE			
RANGE INTLK 1 SWDR AIDL -9B			
ALT HOLD	10.62 ± 1.78 VDC	wl2P2-h	
IN RANGE	13.15 ∓ 1.95 VDC	W12P2-h	
BREAKAWAY	2.82 ∓ 1.16 VDC	W12P2-h	
SWDR AIM-9D/G			
IN RANGE	32,15 ± 75 VDC	W12P2-h	1 MEG
BREAKAWAY	4.11 ± 1.26 VDC	W12P2+h	1 MEG
RANGE INTIK 2. SWDR AIM-9B			
IN RANGE	15,4 ± 2.6 VDC	W12P2-h	
BREAKAWAY	6.295 ± 0.855 VDC	W12P2-h	
SWDR AIM-9D/G			
IN RANGE	27.0 ± 3.5 VDC	W12P2-h	
BREAKAWAY	5.91 ∓ 1.0 VDC	W12P2-h	
3.6.2.3.4			
AIM-7 ( ) RANGE			
	67.00 ± 5.02 VDC	W12P2-h	1 MEG
	61.00 ∓ 7.60 VDC	W12P2-h	1 MEG
	10.20 ∓ 1.33 VDC	W12P2-h	1 MEG
AMI-7D			
ALT HOLD	65.51 ± 4.91 VDC	W12P2-h	1 MEG
IN RANGE	25.50 ∓ 3.80 VDC	w12P2-h	1 MEG
BREAKAWAY	6.10 ∓ 0.98 VDC	W12P2-h	1 MEG
RANGE INTLK 2 AIM-7 E/F			
IN RANGE	33.89 ± 5.05 VDC	W12P2-h	
BREAKAWAY	6.78 ∓ 0.98 VDC	W12P2-h	
AIM-7D			
IN RANGE	16.90 ± 2.62 VDC	W12P2-h	1 MEG
BREAKAWAY	3.389 ∓ 0.98 VDC	W12P2-h	1 MEG

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3.6.2.2.5

RA 4 RMIN STROBES

BREAKSTROBE	25.42 ± 4.79 VDC	W13P1-3	1 MEG
RMIN STROBE	5.083 ± 1.74 VDC	W12P2-3	1 MEG

3.6.2.3.6

FIGHTER MODE

RANGE INTLK 2			
FIRE INTLK LIGHT ON	20.0 ± 2.2 VDC	W12P2-h	1 MEG
FIRE INTLK LIGHT OUT	0 ± .4 VAC	W5P1-n	100 K

3.6.2.3.7

SWEEP CONTROL

SWEEP CONTROL	-6.00 ± 1.62 VDC	W5P1-g	100 K

3.6.2.3.8

ALLOWABLE STEERING ERROR CIRCLE

STEER CIRCLE			
SWDR AIM-0B/D/G	1.400 ± 0.28 VDC	W131-f	100 K
SPARROW AIM-7E/F	1.5337.98 VDC	W13P1-f	100 K
SPARROW AIM-7D	1. 1.689 ± VDC	W13P1-f	100 K

3.6.2.3.9

AIM ( ) ANGLE INTERLOCK

SWDR ANGLE INTLK	-2.1-0.7 VDC	W12P2-A	100 K

3.6.2.3.10

AIM DOT

AIM DOT AZ			
AIM-7E/F	2.03 ± 0.39 VAC RMS	W13P1-F	100 K
AIM-7D	. 73 ± 0.59 VAC RMS	W13PI-F	100 K
AIM DOT EL			
AIM-7E/F	2.03 ± 0.59 VAC RMS	W13P1-g	100 K
AIM-7D	.73 ± 0.50 VAC RMS	W23P1-g	100 K

3.3.2.3.11

VERTICAL GYRO ERECTION CUT-OUT

GYRO IN-GYRO RATE (SWEEP)	LIGHT ON ≥ 30 SEC	W12P2-AG	10 K
GYRO OUT- CYRO RATE (SWEEP)	LIGHT ON ≤ 30 SEC	W12P2-AG	10 K

3.6.2.3.12

LEAD ANGLE PROBLEM AHEAD-AMI, ENGLISH BIAS)

READ AIM AZ			
LW & RW	-39.02 ± 3.21 VDC	W5P1-K)	130 K in parallel
LA	-15.66 ± 3.21 VDG	W5P1-k)	with .022 uf
RA	-32.84 ± 3.21 VDC	W5P1-k)	
LF	-13.41 ± 3.21 VDC	W5P1-k)	
RF	-33.25 ± 3.21 VDC	W5P1-k)	

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HEAD AIM EL			
LW& RW	-13.6 ± 3.21 VDC	W5P1-g & m	130 K in parrallel with .002 uf
LA	-35.53 ∓ 3.21 VDC	W5P1-g & m	
RA	+20.27 ∓ 3.21 VDC	W5P1-g & m	
LF	-35.93 ∓ 3.21 VDC	W5P1-g & m	
RF	+22.52 ∓ 3.21 VDC	W5P1-g & m	
ENG. BIAS-AZ			
LW & RW	-22. 20 ± 5.0 VDC	W5P1-a	240 K in parallel with 160 of
LA	-24.02 ∓ 5.0 VDC	W5P1-a	
RA	-17.43 ∓ 5.0 VDC	W5P1-a	
LF	-24.02 ∓ 5.0 VDC	W5P1-a	
RF	-17.42 ∓ 5.0 VDC	W5P1-a	
ENG. BIAS EL			
LW & RW	+1.5.03 ± 5.0 VDC	WSP1-b	242 Kin parallel with 160 of
LA	+ 7.13 ∓ 5.0 VDC	W3P1-b	
RA	+24.21 ∓ 5.0 VDC	W3PI-b	
LF	+7.13 ∓ 5.0 VDC	W5P1-b	
RF	+24.21 ∓ 5.0 VDC	WSP1-b	

3.6.2.3.13

ALTITUDE COMMANDS

Ogs>5°			
ALT 2	-27.53 ± 3.22 VDC	W12P2-U	1 MEG
ALT 1	-56.40 ∓ 3.64 VDC	W12P2-U	1 MEG
$-5^{\circ} \leq Ogs \leq +5$			
ALT 2	-41.27 ± 3.51 VDC	W12P2-U	1 MEG
ALT 1	-74.58 ∓ 3. 64 VDC	W12P3-U	1 MEG
$Ogs < -5^{\circ}$			
ALT 2 LAMP LIT	-41.274$1 VDC	W12P2-U	1 MEG
ALT 2 LAMP LIT	-56.40 ± 3.64 VDC	W12P2-U	1 MEG
ALT 1	-36.22 ∓ 3.64 VDC	W12P2-U	1 MEG

3.6.2.%14

TARGET INTERCEPT (DOGFIGHT) COMPUTER CHECK

FIRE/ACM MODE			
RANGE/NTLK 2			
AIM-7( )			
IN RANGE	19.58 ± 1.98 VDC	W12P2-b	1 MEG
BREAKAWAY	6.9 ∓ 1.2 VDC	W12P2-b	1 MEG
AIM-9 ( )			
IN RANGE	9.3341.5 VDC	W12P2-b	1 MEG
BREAKAWAY	1.574.24 VDC	W12P2-b	1 MEG
AIM-7 ( )			
RMAX STROBE	14.68$1.48 VDC	W13P1-S	1 MEG
RMIN STROBE	5.15 ± 0.16 VDC	W12P2-g	1 MEG
AIM DOT			
RANGE TO MISSILE	2.65 ± 0.35 VAC RM	W5P1-k	130 K in parallel with .022 of
TAS TO MISSILE	3.54 ± 0.35 VAC RMS	W5Pl-s	130 K in parallel with .022 of
*DC MODE			
RANGE INTLK 2			
AIM-9 D/G (MCP TO HEAT)			
IN RANGE	19.58 ± 1.98 VDC	W12P2-h	1 MEG
BREAKAWAY	1.59 ∓ 0.35 VDC	W12P2-h	1 MEG
AIM-9 D/G (MCP TO HEAT)			
RMAX STROBE	8.9 ± 1.0 VDC	W13P1-S	1 MEG
RMIN STROBE	1.2 ∓ 0.18 VDC	W13P2-g	1 MEG

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AIM-9 D/C (MCP TO RADAR)			
RMAX STROBE	30.0 ± 1.1 VDC	W13P1-3	1 MEG
RMIN STROBE	11.0 ± 0.87VDC	W12P2-g	1 MEG
AIM-9 D/C(MCP TO HEAT)			
RMIN STROBE	11.0 ± 0.37 VDC	W12P2-g	1 MEG
RMAX STROBE	30.0 ± 1.1 VDC	W13P1-S	1 MEG

3.6.2.3.15

MISSILE TUNING

LMI TUNE DOWN	TUNE DOWN LIT IN 16.5 ± 3.5 SEC	W5PI-H	10 K
LIM TUNE UP	TUNE UP LIT	W5P1-H	SO K
RECY & AFC	TUNE DOWN LIT	W5P1-H	10 K
RECY TUNED	DOWN DIM	W5P1-H	10 K
RECY - AFC	TUNE UP LIT	W5PI-H	10 K
RECY NOT TUNED	TUNED	W5P1-H	10 K
AIM-7F LIM	TUNE DOWN LIT IN 16.5 ± 3.5 SEC	W5P1-H	10 K

3.6.23.16

ROLL COMMAND

LEFT & RIGHT WING	0 ± 3.1 VDC	W 5P1-c	1 MEG
LEFT FUSELAGE	-18.8 ± 3.1 VDC	W5P1-c	1 MEG
RIGHT FUSELAGE	+18.8 ± 3.1 VDC	W5P1-c	1 MEG

3.6.2.3.17

AMI-7 ( ) MISSILE POWER

115 VAC 400HZ / A	115 ± 11.5 VAC RMS	W5P1-D	1 MEG
115VAC 400HZ / B	115 ± 11.5 VAC RMS	W5P1-E	1 MEG
115VAC 400HZ / C	115 ± 115 VAC RMS	W5P1-C	1 MEG
			-LOADED WITH 68 n <5 SECONDS
TIMING+POWER CUT OFF	1.2 ± .1 SEC	W5P1-S & W5P1-D	50 K
TIMING. 5 SECOND DELAY	0.5 ± .05 SEC	W5P1-S & W5P1-D	50 K
TIMING-2 SECOND DELAY	2.0 ± .36 SEC	W5PI-5	60K

3.6.2.3.18

AIM-7 ( ) MOTOR FIRE & AIRCRAFT INTERLOCKS

TIMING - MOTOR FIRE	1.3 ± .2 SEC	W5P1-S & W5P1-h	50 K LOADED WITH 4 -<5 SECONDS

3.6.2.3.19

SWEEP SELECT & GATE/ASPECT

TIMING - POWER CUTOFF	0.3 ± .05 sEc	W5P1-S & W5P1-D	SO K

3.6.2.3.20

AIR DATA COMPUTER CHECK

TAS 150V	53. %8.1 VDC	W12P2- d	1 MEG
ALI DC 150V	-53.0*8. 5 VDC	W12P2-Z	1 MEG
ALI POT 150V	-54.0 + 8. 5 VDC	W12P2-L	1 MEG
ATTACK ANGLE 60V	0 ± 2 VDC	W12P2-a	1 MEG
AOA (25 UNITS)	19.0 ± 5 VAC RMS	W12P2-a	1 MEG
AOA (6 UNITS)	0.0 ± 5 VAC RMS	W12P2-a	1 MEG
AOA (1 UNIT)	5.0 ± 5 VAC RMS	W12P2-a	1 MEG

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3.6.2.3.21 The following wires are capable of switching the indicated currents at the proper program voltages.

CABLE W5

P1	P2		Current (MA)
R	f	+28 VDC MOTOR	50
d	v	MOTOR TUNER AFC	5
e	s	RECYCLE	5
f	T	KLYSTRON LIMIT	5

CABLE W12

P2	P1 (Jl12, 73-J404)		
A	J	EL LOS RATE SIGNAL TO COMPUTER	5
C	p	+28 VDC RANGE LOCKON (TO COMPUTER)	50
O	a	AZ LOS RATE SIGNAL TO COMPUTER	5
T	d	+28 VDC ANGLE LOCK	50
U	AF	ALT DC RANGE INTLK CKTS	5
h	u	RANGE VOLTAGE (TO COMPUTER)	5
j	w	TRUE AIR SPEED (TAS)	5
n	E	V_c IN	100
v	AA	+28 VDC ACM MODE	50
x	G	ALT AC TO COMPUTER	100
AF	m	AIM-7E MISSLE SELECT (TEST)	50

CABLE W13

P1	P2 (J113, 73-J405)		
B	f	+28 VDC BREAK	50
K	K	+28 VDC TIME DELAY OVERRIDE (2 SEC)	500
L	R	+28 VDC FROM NC CONTACTS OF MGR	500
P	P	+28 VDC STBY	500

SIGNALS FROM RADAR ON CABLE W12

P2	P1 [J112. 73-J404)		Current (MA)
G	R	EL LOS RATE SIGNAL	5
K	x	+28 VDC RANGE LOCKON	50
N	U	+28 VDC LOCKON To INDICATOR CONTROL	50
Y	k	AZ LOS RATE SIGNAL	5
C	M	V_c OUT	100
	AC	RANGE VOLTA GE	5

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3.6.3 Electrical Cable Test Set TS-3648/AWM-87 -Capability shall be provided to verify the integrity of the multi-pin cables W5, W12 and W13. Verification shall consist of locating an open, short circuit to ground or a short toadjacent wires. A NO-GO indication shall be provided if one or more faults are detected and the faulty pin identification shall be digitally displayed to the operator. A Target Simulator shall be included within the TS-3648/AWM-87 to provide simulated video target signals to the AMCS. The targets shall be at fixed ranges, velocities and accelerations as defined below.

3.6.3.1 Built-In Test - The equipment shall contain integral circuitry activated by a front panel control that generates the necessary test signals and references to enable end-to-end functional checks. The confidence level of this built-in test shall be 90% minimum.

3.6.3.2 Target Simulator - A simulated video target signal shall be available for radar system test applications. The simulator shall have the following modes of operation:

3.6.3.2.1 Initial Range - Capability of setting initial range of a target at 5 and 40 nautical miles, with an accuracy of +l%

3.6.3.2.2 Range Rate - Capability of generating a constant velocity target, opening at 600 knots with an initial of 5 and 40 nautical miles and closing at 600, 1600, and 2500 knots with an initial range of 40 nautical miles and with an accuracy of ± 1%

3.6.3.2.3 Acceleration - Capable of accelerating a target with initial range of 40 nautical miles initial range rate 100 knots (closing) at 1.6, 2.4, and 5g's with an accuracy of + 1%

3.6.4 Protection - The equipment shall be capable of withstanding input power variations as imposed by abnormal electric system operations as defined in ground MIL-STD-704 and input signals twice normal.

3.6.5 Warm-Up Time - The time required for the equipment to warm-up prior to operations shall be kept to a minimum and shall not exceed 10 minutes under standard conditions and 20 minutes at extreme service conditions.

3.7 Detail Requirements

3.7.1 Cable Set, Interconnecting Cable Group - The cable set shall meet the following requirements:

3.7.1.1 Length

W5	38' ± 1'
W12	8' ± 1/2'
W13	8' ± 1/2'

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3.7.1.2 Terminations

W5P1	MS3106E28-21S
W5P2	Kern Engineering CT0699 with Amphenol insert 115-1388
W5P3	Alden Products #6102 UL51
W12Pl	MS3106E32-2O1O-1OP
W12P2	MS3106E32-2O1O-1OS
W13Pl	MS3106E32-2O1O-1OSW
W13P2	Amphenol 67-06E22-48P

3.7.1.3 Shielding - The cables shall be shielded such that no RF pickup shall be conducted into the Fire Control System Test Set circuitry.

3.7.1.4 Cable Set Container - The cable set shall include a combination case for their storage. The overall dimensions of this container shall not exceed 14 inches wide by 14 inches high by 21 inches long.

3.7.1.5 Weight - The total weight of cable set and container shall not exceed 45 pounds.

3.7.2 Test Set, Fire Control System TS-3649/AWM-87 - The test set shall meet the following requirements:

3.7.2.1 Form Factor - The overall dimensions of the test set shall not exceed 14.5 inches wide by 12.75 inches high by 21 inches long.

3.7.2.2 Weight - The weight of the test set shall not exceed 55 pounds.

3.7.2.3 Content - The test set shall include a power cable, circuit card extender hoard, card extractor and power cable adapter stored in the cover.

3.7.2.4 External Connectors - External connectors shall provide for the following:

Reference Designation	Connector Type	Function
Jl (W5P1)	MS3102R28-2IP	Interface with missile launch stations
J2 (W12P2)	MS3102R32-2O1O-1OP	Interface with radar access panel connectors
J3 (W13Pl)	MS3102R32-2O1O-1OPW	
J4	MS3102A-2O-4P	Primary power in

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3.7.2.5 Controls - Controls shall be provided for the following functions:

Control	Function
ON-OFF (SW)	Control primary power (115 vAC, 400 Hz)
TEST SELECTOR (Rotary)	Select one of 15 system test series plus self test: SELF CHECK SWEEP SELECT AIM-7 MTR FIRE AIM-7 MSL PWR MISSILE TUNING LEAD ANGLE ALT COMMAND RANGE AIM-7 RANGE AIM-9 FIGHTER MODE SWP CONT, ERROR CIR, ANGLE LNTLK, AIM DOT FIRING INTERLOCK GYRO ERECTION COMPUTER CHECK INTERCEPT (DOGFIGHT)
TEST STEP (S.w)	Individual test stepping sw
OPERATOR ACTION (SW)	Verification of action taken in aircraft
OPERATOR OBSV (SW)	Verification of required observation in aircraft
MOTOR FIRE (Momentary Sw)	Apply load to motor fire circuitry
MISSILE SELECT (Rotary)	Select appropriate missile station
PARITY (Sw)	Activates self test, lamp test or normal operation
CB (Resetable circuit breaker)	Protect the test set and input line from overload

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3.7.2.5 Controls - Controls shall be provided for the following functions:

Control	Function
ON-OFF (Sw)	Control primary power (115 VAC, 400 Hz)
TEST SELECTOR (Rotary)	Select one of 15 system test series plus self test: SELF CHECK SWEEP SELECT AIM-7 MTR FIRE AIM-7 MSL PWR MISSILE TUNING LEAD ANGLE ALT COMMAND RANGE AIM-7 RANGE AIM-9 FIGHTER MODE SWP CONT, ERROR CIR, ANGLE INTLK, AIM DOT FIRING INTERLOCK GYRO ERECTION COMPUTER CHECK INTERCEPT (DOGFIGHT)
TEST STEP (S.W)	Individual test stepping sw
OPERATOR ACTION (SW)	Verification of action taken in aircraft
OPERATOR OBSV (SW)	Verification of required observation in aircraft
MOTOR FIRE (Momentary Sw)	Apply load to motor fire circuitry
MISSILE SELECT (Rotary)	Select appropriate missile station
PARITY {SW)	Activates self test, lamp test or normal operation
CB (Resetable circuit breaker)	Protect the test set and input line from overload

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LAMP TEST (Momentary SW)	Test lamps
TEST MODE(Rotary)-Target Gen	Selects initial range, range rate and target OFF
TEST SELECT (Rotary)- Target Gen	Select target dynamics

3.7.3.6 Status Indicators - The following status lights, indicators and readouts shall be provided:

Display	Function
OUTPUT	Output Trigger
POWER	Primary power ON
GO/NO-GO	Indicate cable condition
STEP NO.	Pin number indication

3.9 Finish - Front panel finish will be black anodized per MIL-A-8625, black, Class 2, Type 2.

3.10 Interchangeability - Interchangeability at major module level shall be in accordance with MIL-STD-454, requirement 7.

3.11 Workmanship - Workmanship shall be in accordance with MTL-STD-454, requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 General - The test equipment shall be subjected to the following tests to determine compliance with all the applicable requirements.

a. Preproduction Tests
b. Acceptance Tests
c. Environmental Qualification Tests

4.2 Preproduction Tests - Preproduction tests shall be made on one or more test equipments representative of the production equipments to be supplied under the contract. Pre-production tests shall consist of the following:

a. Demonstration tests
b. Fleet approval tests

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4.2.1 Demonstration Tests - Contractors demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with an approved test procedure of paragraph 4.4. Notification when tests are to be conducted so that a representative may be designated to witness the tests when so desired shall be required.

4.2.2 Fleet Approval Tests - At the completion of the Contractor's demonstration tests and when specified by the contract the equipment shall be delivered to the fleet for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.3 Acceptance Tests - The Contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection anti testing shall be under the supervision of the procuring activity. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

a. Individual Tests
b. Special Tests

4.3.1 Individual Tests - Each equipment submitted for acceptance shall be rejected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship and operational adequacy. As a minimum, each equipment accepted shall have passed the following tests:

a. Examination of Product
b. Operation Test

4.3.1.1 Examination of Product - Each equipment shall he examined carefully to determine that the material, workmanship and interchangeability requirements have been met.

4.3.1.2 Operation Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure compliance with the design requirements of this specification.

4.3.2 Special Tests - Special tests as defined by the contract shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. These tests shall be subject to the clause of the contract entitled "CHANGES".

4.3.2.1 Special Test Schedule - Selection of equipments for special test shall be made as follows:

On the first equipment after an engineering change.
When failure reports or other information indicate additional equipment should be tested.

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3.7.2.6 Status Indicators - The Test Set shall have the following status indicators and readouts:

Display	Function
BRK AWAY HYD ACT FUZE GND GYRO IN SWP SEL HD & LCH IN RANGE ALT HOLD SW AFT BOOST MTR FIRE EPU ALT 1 ALT 2 TUNE UP TUNE DOWN TUNED PHASE FIRE INTLK	Aircraft status
OPERATOR ACTION OPERATOR OBSV	Indicate need fur action in aircraft cockpit
Go/NO-Go	Test result
TEST VALUE	Digital readout of test result
TEST STEP	Number of tests to be initiated
PWR ON	Primary power ON
ACFT PWR	28 V from aircraft ON
MISSILE POWER	Motor fire control ON
FAULT/SELF TEST	Indicates parity fault or self test status
MISSILE POSITION LAMPS	Indicates one of six missile stations under test
TEST SELECTOR LAMPS	Indicates System Test

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3.7.3 Electrical Cable Test Set TS-3648/AWM-87 - The Test Set shall meet the following requirements.

3.7.3.1 Form Factor - The overall dimensions of the test set shall not exceed 14.5 inches wide by 12.75 inches high by 23.25 inches long.

3.7.3.2 Weight - The weight of the test set shall not exceed 45 pounds.

3.7.3.3 Content - The test set shall include a power cable, circuit card extender board, card extractor and power cable adapter stored in the cover.

3.7,3.4 External Connectors - External connectors shall provide for the following:

Reference Designation	Connector Type	Function
J1 (W12P1)	MS3102R32-2O1O-1O5	
J2 (W13P1)	MS3102R32-2O1O-10PW	Connection for cable to be tested
J3 (W12P2)	MS3102R32-2O1O-10P	
J4 (W5P3)	Alden 6102 ULSI	
J5 (W5P1)	MS3102R28-21P	
J6 (W13P2)	67-02E22-485	
J7 (W5P2)	Amphenol 1328	
J8	MS3102A-2O-4P	Primary power
J9 (In)	BNC-F	Trigger In
J10 (Out)	BNC-F	Target Out

3.7.3.5 Controls - Controls shall be provided for the following functions:

Control	Function
POWER ON - OFF (Sw)	Control primary power (115 VAC, 400 Hz)
CABLE SELECT (Rotary)	Select W5, W12, W13 cable or self test
AUTO-MANUAL (Sw)	Select Automatic or manual sequencing of pin checks
INCREMENT (SW)	Step through pins in manual mode and initiates auto stepping in auto mode
CLEAR/NO-GO (Momentary SW)	Allow test to continue after NO-GO indication

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4.3.2.2 Scope of Test - Special tests shall consist of such tests as are selected by the procuring activity. Only such tests that are required to check the characteristics in question need be conducted.

4.3.3 Equipment Failures - Should a failure occur during any tests of Section 4.0 the following action shall be taken:

a. Determine the cause of failure.
b. Determine if the failure is an isolated case or design defect.
c. Where practical, include a test in the individual test to check all equipment for this requirement until assurance is obtained that the defect has been satisfactorily corrected.

4.4 Test Procedure - Preproduction test, acceptance tests and environmental qualification tests shall be performed in accordance with a test plan shall be prepared by the contractor. MLL-T-18303 shall be used as a guide for preparation of test procedures.

4.5 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and may be resubmitted for unit acceptance.

4.6 Environmental Qualification Test - The contractor shall perform an environmental qualification test. Test limits, levels and other qualifying conditions shall be as specified in paragraphs 3.4.5 and 3.4.6 of this specification.

5. PREPARATION FOR DELIVERY

5.1 General - All units and parts of equipment shall be preserved, packaged, packed, and marked for the level of shipment specified in the contract or in accordance with MIL-E-17555.

6. NOTES

6.1 Intended Use - These equipments shall be designed as portable units housed in combination cases. Unit dimensions, weight and handles shall be consistent with intended usage as portable PGSE for organizational maintenance applications.

6.2 Source Data- The technical content of this specification has been established by information contained in the following unclassified sources of data:

NAVAIR 01-245FDN-2.8.5 Technical Manual
NAVAIR 16-30APQ72-6-1 Technical Manual

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6.3 Auxiliary Ground Support Equipment - The following auxiliary ground support equipment shall be required:

a. AN/USM- 180 Oscilloscope or equivalent
b. HP3450B Digital Voltmeter or equivalent
c. AN/AWM-52A Launcher Ejection Test Set
d. MSTS
e. TS1736/AWM-8 RF Test Set
f. AN/APM-383 Radar Test Set

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-T-85068(AS) Test Set, Fire Control System AN/AWM-87

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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